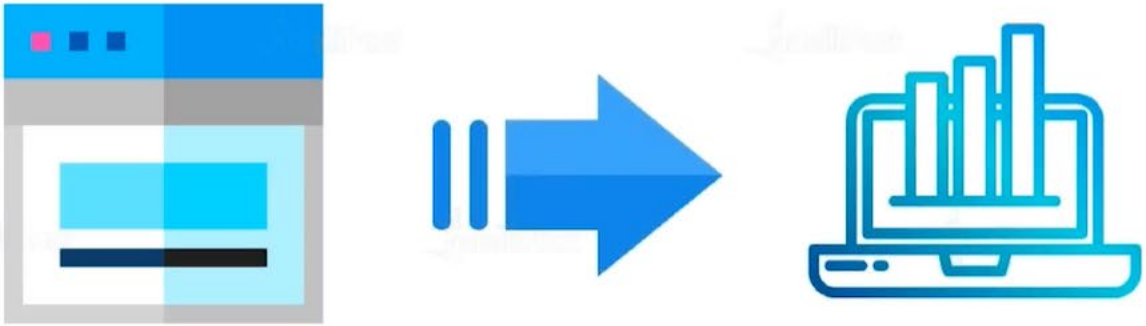
What is Web Scraping?

Web scraping is the process by which we extract useful data from a web page on the Internet

Web scraping is done by parsing the HTML content of a website and then navigating to the desired content and extracting it from the web page



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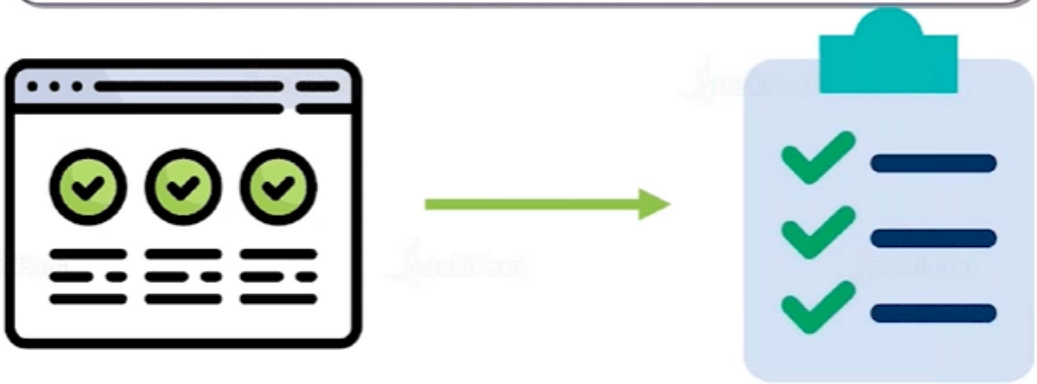
Why use Web Scraping?

You can use Web Scraping in many different fields

In lead generation, market analysis, and in our case collecting and testing data set for Machine Learning. For example, contact details of businesses as well as individuals from yellow pages websites



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What is HTML?

HTML (Hypertext Markup Language) is a markup language used to describe the structure and content of a web page

HTML contains data in the from of text encapsulated within tags and attributes, and this information is what we obtain through scraping a web page

```
<a href="http://www.myblog.com/sample_post">Link to my sample blog post</a>
```

Tag | attribute | attribute value | content

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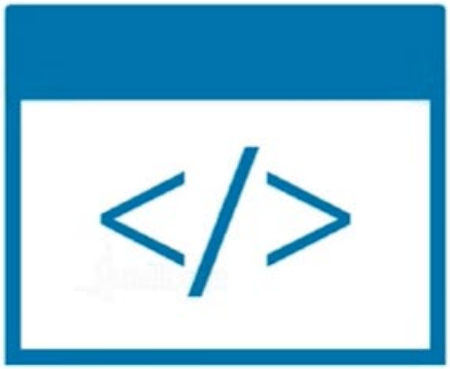
HTML Tags

In HTML, tags are the basic building blocks of the structure of a web page

A tag helps a web browser decide what an element on a web page is, e.g., an image, a paragraph, etc.



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HTML Attributes

HTML attributes are added to HTML tags to provide additional information

Attributes come in key–value pairs, e.g., alt="sample image"

HTML Selectors

In HTML, if we wish to refer to a particular element on the web page, we need to specify some sort of selector on that element

A selector is basically an identifier that is put on the HTML elements so that they can be programmatically accessed later

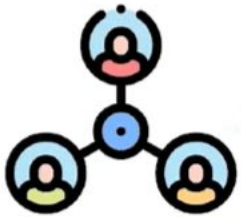
HTML Selectors

There are two types of selectors available in HTML

ID

An ID is used as a selector when we wish to identify a single element on a web page. An ID can only be assigned to a single element on the page

Class



HTML DOM

When HTML is parsed, it is converted into a hierarchical tree structure

This tree structure is called Document Object Model (DOM). This tree structure is what we use when we access an HTML-element on the web page

```
<div>
  <a href="jane">Jane</a>
  <a href="john">John</a>
</div>
```

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Beautiful Soup

Beautiful Soup is a Python package that is used to scrape data from web pages

It uses parsing libraries to parse the HTML content of a web page that can then be used to extract contents from that web page

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